

036015

Finite Elements Method for Engineering Analysis

Homework Assignment 3 - Coded 1D Linear Element

Winter 2026

Instructions

- Electronic submission of **typed** sheets only (PDF format through moodle).
- Submission file must be named: "*name surname.pdf*". No I.D number or homework number is required in the file name.
- Solution must contain all the relevant derivations, explanations, graphs and conclusions.
- All codes used must be added to the submission files.

Consider an elastic rod, fixed to a rotating hinge as shown in Figure 1:

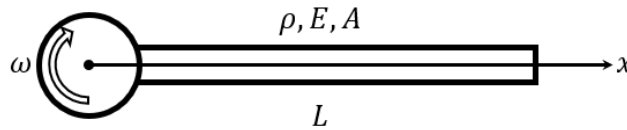


Figure 1: Uni-axially loaded rotating bar

where $E = 200\text{GPa}$ is Young's modulus, $A = 15\text{cm}^2$ is the constant cross section area of the rod, $L = 2\text{m}$ is the length of the rod, $\rho = 7850\text{Kg/m}^3$ is the constant density of the rod and $\omega = 2\pi\text{rad/sec}$ is the angular velocity of the rotation.

Neglecting the bending of the rod, the BVP that governs the problem is given by:

$$\begin{cases} -\frac{d}{dx} \left(EA \frac{du}{dx} \right) = \rho A \omega^2 x \\ u(0) = 0 \\ EA \frac{du}{dx} \Big|_{x=L} = 0 \end{cases} \quad (1)$$

Where $u(x)$ is the stretch of the rod along the x axis during rotation as a result of the apparent outward force - the centrifugal force.

The MATLAB script for the exact solution (copy and paste in script):

```
u_exact = -(r*w^2)/(2*E)*((1/3)*x.^3-L^2*x)
```

The following part is coded:

1. Write a Matlab code to solve the problem for an arbitrary number of linear elements n . Make sure your code is properly documented. Use the local matrices that you have derived in the previous home work assignment. There is no need to show the derivation again. [75pt]

Tip: Look at the "1D matlab examples" file for inspiration. You will be graded on the accuracy and readability of your code, as well as demonstrating your understanding of



the method. Copy-pasting from the example file will not garner points unless each line is properly documented.

2. Plot the exact solution, as well as the approximated solution using 1, 3, 100 elements on the same graph. [25pt]

Tip: Only plot the values of the nodal points connected with linear lines (default `plot` function in Matlab). No need to draw the interpolation using the basis functions for this assignment.

3. Bonus [20pt]: Plot a **logarithmic** (Matlab function: `loglog(...)`) graph of the L_2 and H^1 norms of the error vs. the **length of a single element** h for 1, 5, 10, 50, 100, 500, 1000, 2000 elements.

Tip: look at "1D matlab examples" file for inspiration.

Tip: calculations of the integrals in local coordinates of the elements requires the Jacobian.

Calculate the norm integral using Gauss quadrature for each element. Calculate the slope of the norm convergence graph, and print it on the figure.

For your use:

$$L_2 = \sqrt{\int_{\Omega} (u - \tilde{u})^2 dx} \quad H^1 = \sqrt{\int_{\Omega} (u - \tilde{u})^2 + \left(\frac{du}{dx} - \frac{d\tilde{u}}{dx} \right)^2 dx} \quad (2)$$

Calculate norms using Gauss integration as follows:

$$\begin{aligned} L_2 &= \sqrt{\int_{\Omega} (u - \tilde{u})^2 dx} = \sqrt{\underbrace{\int_{\Omega_1} (u - \tilde{u})^2 dx}_{\text{Element 1}} + \underbrace{\int_{\Omega_2} (u - \tilde{u})^2 dx}_{\text{Element 2}} + \dots + \underbrace{\int_{\Omega_n} (u - \tilde{u})^2 dx}_{\text{Element n}}} \\ &= \sqrt{\underbrace{\int_{-1}^1 (u(x_1(\xi)) - \tilde{u}_1(\xi))^2 |J_1| d\xi}_{\text{Element 1}} + \underbrace{\int_{-1}^1 (u(x_2(\xi)) - \tilde{u}_2(\xi))^2 |J_2| d\xi}_{\text{Element 2}} + \dots + \underbrace{\int_{-1}^1 (u(x_n(\xi)) - \tilde{u}_n(\xi))^2 |J_n| d\xi}_{\text{Element n}}} \end{aligned} \quad (3)$$



Substitute the Gauss point $\xi = 0$ and weight $w = 2$:

$$L_2 = \sqrt{2} \left[\underbrace{(u(x_1(0)) - \tilde{u}_1(0))^2 |J_1|}_{\text{Element 1}} + \underbrace{(u(x_2(0)) - \tilde{u}_2(0))^2 |J_2|}_{\text{Element 2}} + \dots + \underbrace{(u(x_n(0)) - \tilde{u}_n(0))^2 |J_n|}_{\text{Element n}} \right]^{1/2} \quad (4)$$

In order to calculate H^1 use 2 Gauss points.

Good Luck!